CS319:

Frontend Development

Week2: *Web Structure With HTML*



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What is HTML?

What Is HTML?

- **HTML** stands for Hyper Text Markup Language
- **HTML** is the standard markup language for creating Web pages
- **HTML** describes the structure of a Web page
- **HTML** consists of a series of elements
- **HTML** elements tell the browser how to display the content
- **HTML** elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

https://www.w3schools.com/html/html_intro.asp

The Organizations Behind HTML

W3C (1994)

(World Wide Web Consortium)

- Founded by Tim Berners-Lee, the inventor of the Web
- An open, international consortium of companies, universities, and organizations
- Publishes standards through a formal review process, producing dated "snapshot" specifications
- Pushed XHTML in the early 2000s and halted HTML development in 1998
- Still maintains CSS, SVG, WCAG, and WAI-ARIA today

WHATWG (2004)

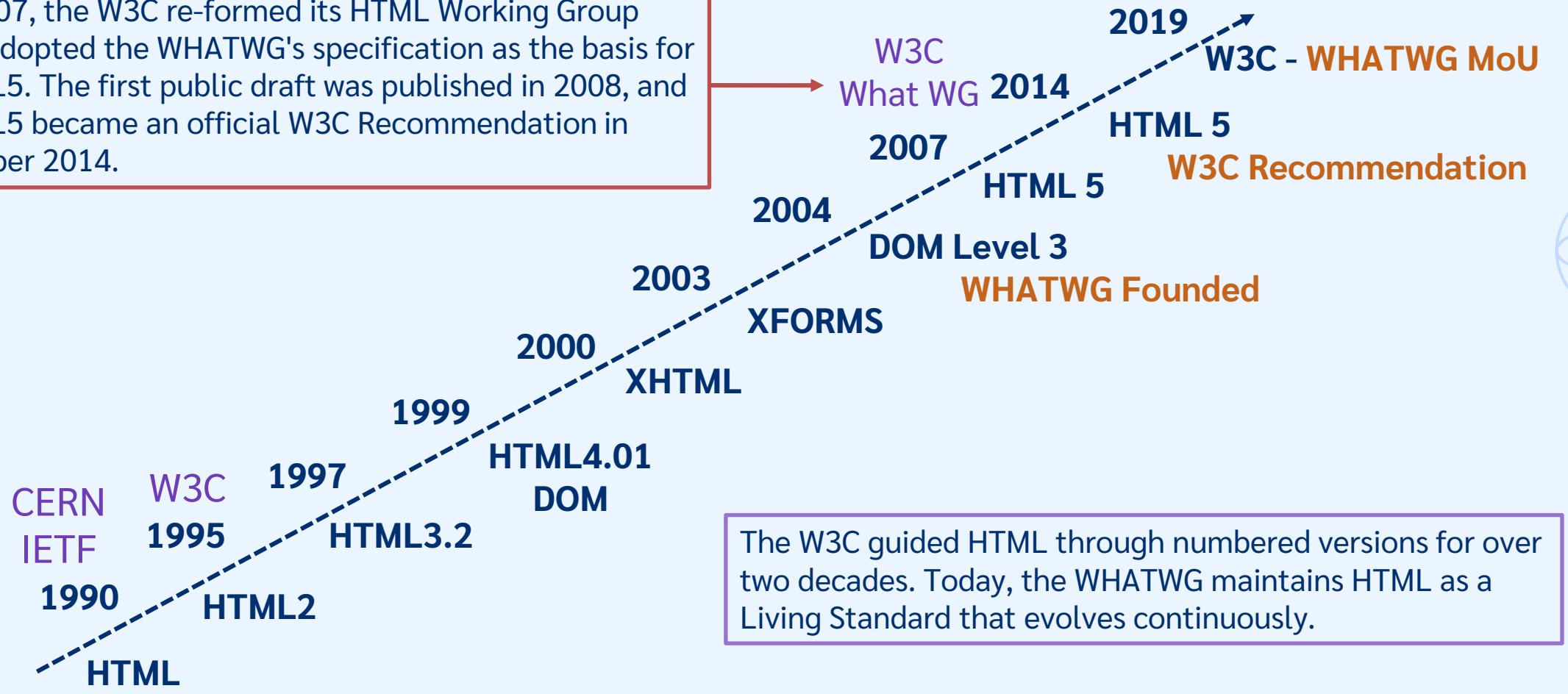
(Web Hypertext Application Technology Working Group)

- Formed by Apple, Mozilla, and Opera after the W3C rejected their proposal to continue evolving HTML
- Believes standards must stay backward compatible and match what browsers implement
- Maintains HTML as a "Living Standard" — continuously updated, with no version numbers
- Google joined later; today the four browser vendors steer the group

html.spec.whatwg.org

The Evolution of HTML - Timeline

In 2007, the W3C re-formed its HTML Working Group and adopted the WHATWG's specification as the basis for HTML5. The first public draft was published in 2008, and HTML5 became an official W3C Recommendation in October 2014.



The W3C guided HTML through numbered versions for over two decades. Today, the WHATWG maintains HTML as a Living Standard that evolves continuously.

HTML vs XHTML

HTML

```
<!-- HTML5 — forgiving -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Page</title>
</head>
<body>
  <P>Hello, world!
  <br>
  <img src="cat.jpg" alt="A cat">
  <input type="checkbox" checked>
</body>
</html>
```

XHTML

```
<!-- XHTML — strict -->
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
  <title>My Page</title>
</head>
<body>
  <p>Hello, world!</p>
  <br />
  <img src="cat.jpg" alt="A cat" />
  <input type="checkbox" checked="checked" />
</body>
</html>
```

XHTML demands perfection. HTML forgives mistakes (Forget a closing tag, and the page still renders). The web is imperfect and browsers must keep it working. So, HTML won.

HTML5 Capabilities

Structure: Semantic Elements (Tags)

- <header> <nav> <main> <section> <article> <aside> <footer>
- Better for accessibility, SEO, and code readability

Media: No Plugins Required

- <video> and <audio> play natively

Forms: Built-in Input Types and Validation

- type="email" "date" "number" "color" "range" "tel"
- Attributes: required, placeholder, pattern, min, max
- Browsers validate before submit

Storage and Offline

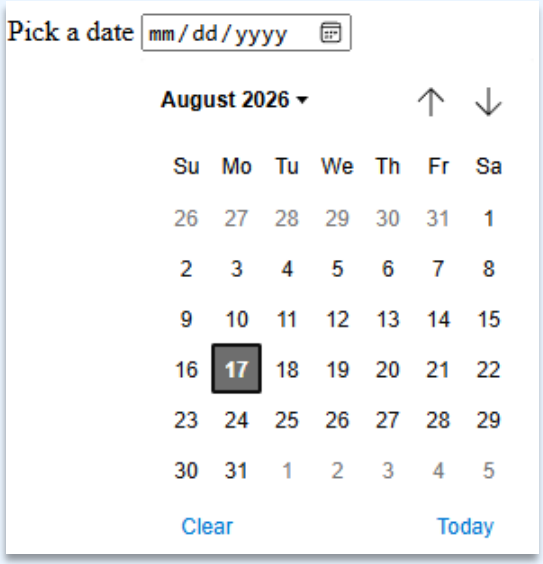
- Service Workers enable offline-capable web apps

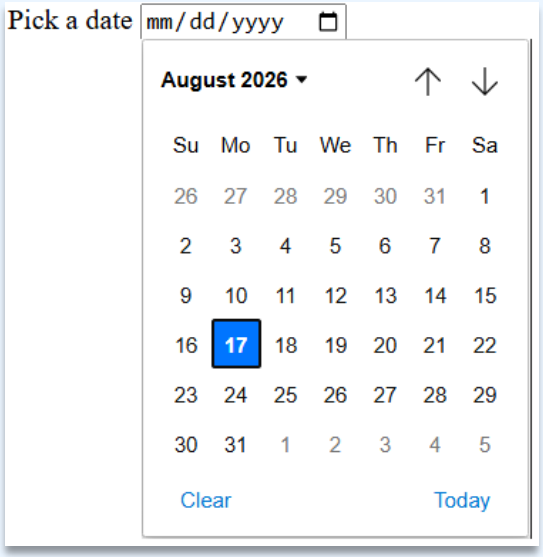
Device and Real-time APIs

- Geolocation API - access the user's location (with permission)
- Drag and Drop API
- Web Workers - run scripts in the background
- WebSockets - real-time two-way communication

HTML5 Browser Support

Some browsers do not fully support all HTML5 and CSS capabilities and their visual may differ.

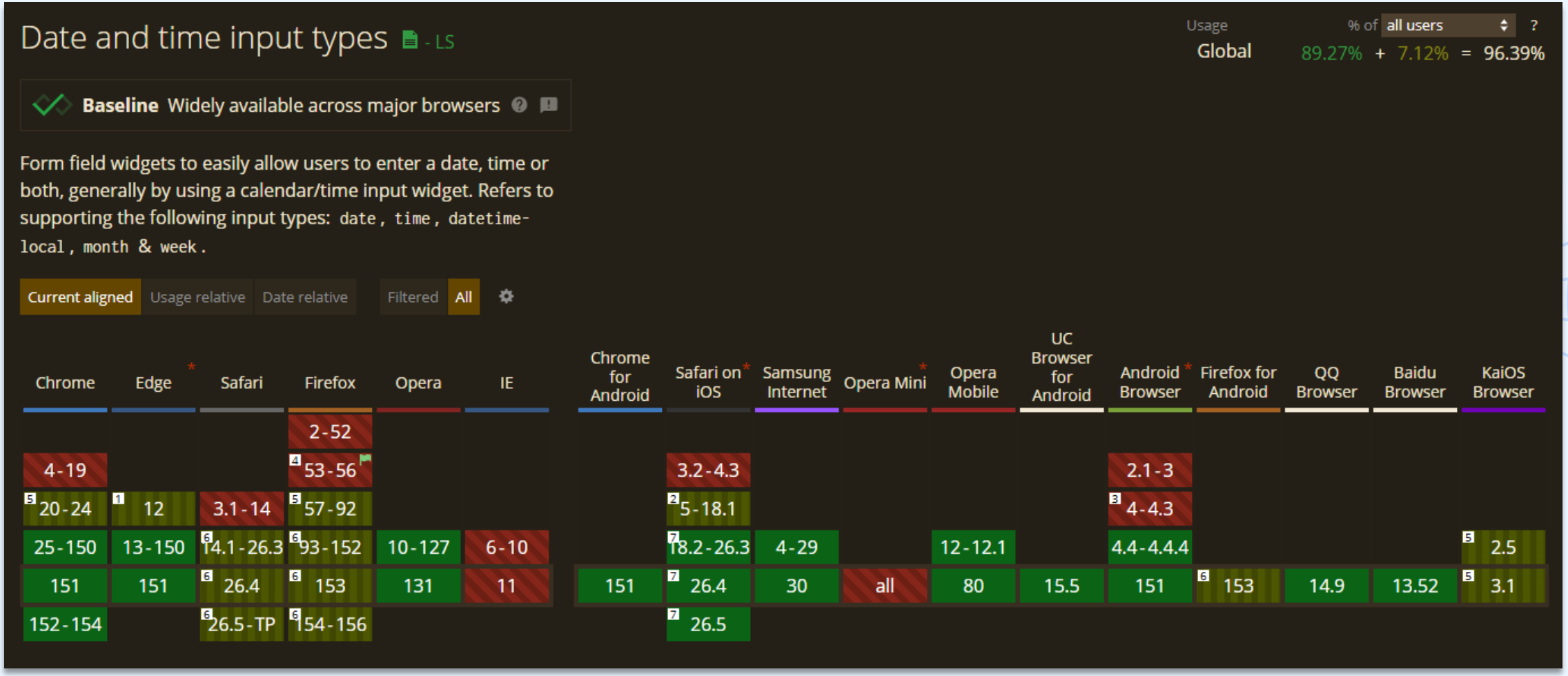






https://www.w3schools.com/tags/ref_html_browsersupport.asp

Checking Browser



https://caniuse.com/



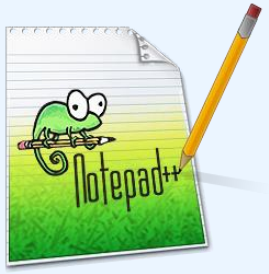
Let's Try HTML!

Open Any Text Editor or IDE

Even Notepad can create a HTML Document!

Offline IDE/Text Editor











Online IDE



W3Schools Spaces

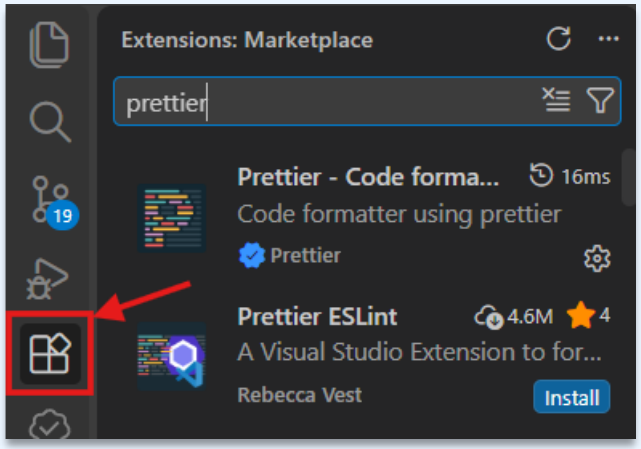
https://www.w3schools.com/spaces/index.php



Code Pen

https://codepen.io/

Useful VSCode Extension & Shortcut

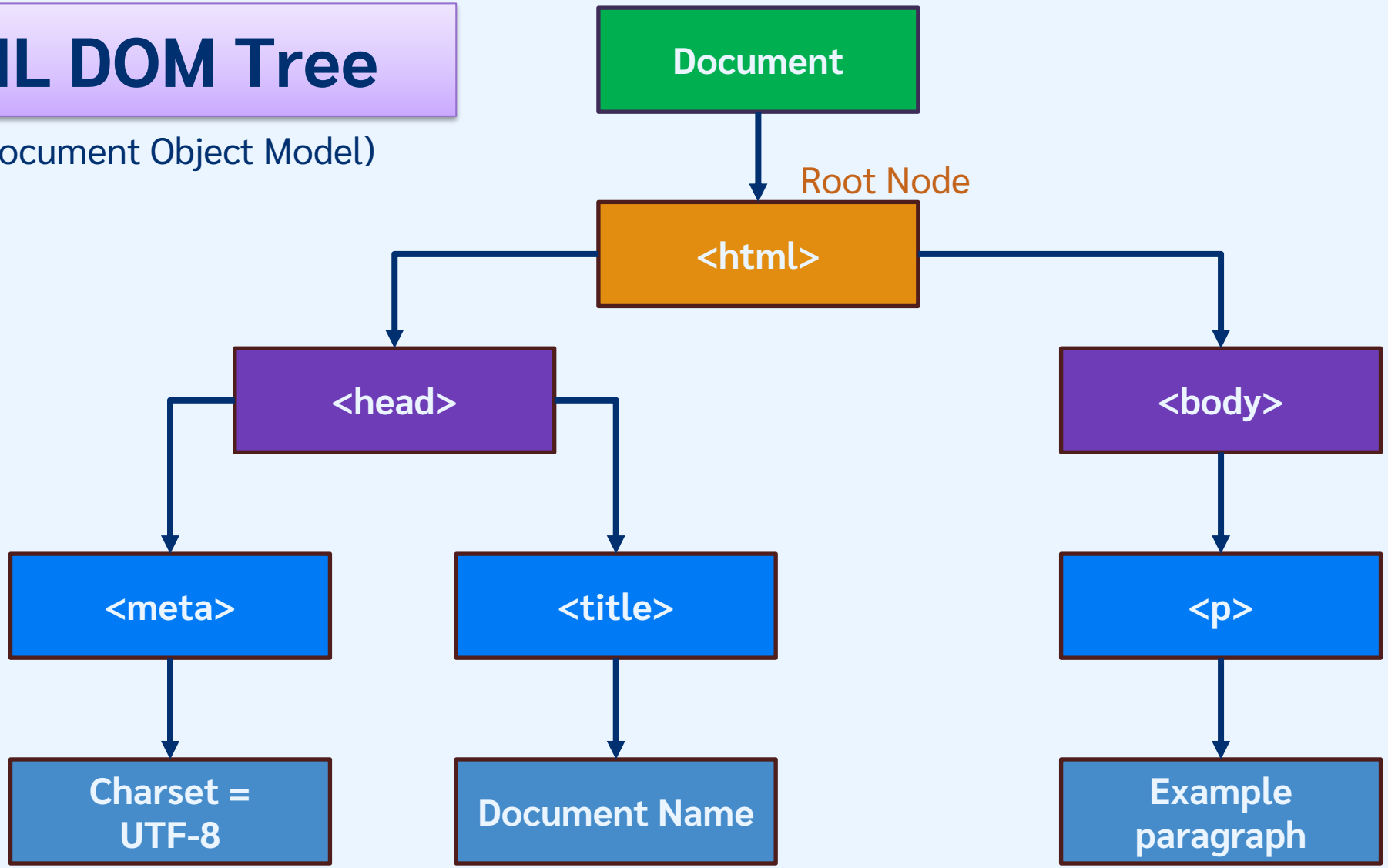


- HTML CSS Support
- Live server
- Prettier - Code formatter
- Auto Rename Tag
- IntelliSense for CSS class names in HTML
- Material Icon Theme

You can press “shift + alt + p” to format your document.

HTML DOM Tree

(DOM: Document Object Model)



Activity - Your First HTML!

Using <tag> to create an element in HTML and close a tag with /

```
<!DOCTYPE html>
<html>
    <head>
        <meta charset="UTF-8">
        <title>Sample Page</title>
        <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    </head>
    <body>
        <h1>Hello World</h1>
        <p>This is my first page.</p>
        <!-- this is a comment -->
    </body>
</html>
```

- **<!DOCTYPE html>** represents the document type
- **<html>** is the root element while **<head>** and **<body>** are its two children
- **<head>** contains metadata and initialize
- **<body>** contains everything visible on the page

https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements

Character Encoding

```
<!DOCTYPE html>
<html>
    <head>
        <meta charset="UTF-8">
        <title>Sample Page</title>
        <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    </head>
    <body>
        <h1>Hello World</h1>
        <p>This is my first page.</p>
        <!-- this is a comment -->
    </body>
</html>
```

- Different languages use different character sets, or charsets. This tag declares which character set is to be used.
- The most common charset used by most languages is UTF-8

Your First HTML! (Typography)

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>
<hr>
<p>Bangkok University was founded in 1962 by Mr. Surat and
Mrs. Pongtip Osathanugrah.It began in the Kluaynamthai area
of Bangkok as a small college..</p>
<p>
    Today the university has two campuses:<br>
    City Campus, Rama 4 Road, Bangkok<br>
    Rangsit Campus, Khlong Luang, Pathum Thani
</p>
<hr>
```

- **<h1>** to **<h6>** headings, largest to smallest
- **<hr>** horizontal divider (no closing tag)
- **<p>** defines a paragraph, auto-spaced
- **
** line break inside a paragraph (no closing tag)

HTML collapses whitespace, extra spaces and line breaks in your code do not appear on the page

Your First HTML! <a>

<a> the anchor element, used to create links

```
<a href="https://www.bu.ac.th/">BU Website</a>
<a href="https://www.bu.ac.th/" target="_blank"> BU Website2</a>
<a href="about.html" target="_blank"> About us</a>
<a href="#top">Top of Page</a>
<a href="#section">Link to section</a>
<br><br><br><br><br><br><br><br>
<h1 id="section">Section</h1>
```

- **href** attribute tells the browser where to go
  "https://www.bu.ac.th/" → another website
  "about.html" → another page in your site
  "#section" → an element on the same page
- The text between **<a>** and **** is what the user clicks
- **target="_blank"** opens the link in a new tab
- **id="section"** marks a destination that #section can jump to

Your First HTML!

<ul> create unorder list, **<ol>** to create order list
and using **<li>** to create a list item

```
<!-- Order does not matter -->
<h2>Things You Need</h2>
<ul>
    <li>A text editor</li>
    <li>A web browser</li>
    <li>Money</li>
</ul>

<!-- Order matters -->
<h2>Making Your First Page</h2>
<ol>
    <li>Create a file named index.html</li>
    <li>Write your HTML</li>
    <li>Save the file</li>
    <li>Open it in your browser</li>
</ol>
```

Your First HTML! <table>

```
<table>
    <tr>
        <th>Heading 1</th>
        <th>Heading 2</th>
        <th>Heading 3</th>
    </tr>
    <tr>
        <td>H1 - Data Item 1</td>
        <td>H2 - Data Item 1</td>
        <td>H3 - Data Item 1</td>
    </tr>
    <tr>
        <td>H1 - Data Item 2</td>
        <td>H2 - Data Item 2</td>
        <td>H3 - Data Item 2</td>
    </tr>
    <tr>
        <td>H1 - Data Item 3</td>
        <td>H2 - Data Item 3</td>
        <td>H3 - Data Item 3</td>
    </tr>
</table>
```

- **<table>** defines the whole table
- **<tr>** table row, one horizontal line of cells
- **<th>** table header cell, bold and centered by default, and read as a header by screen readers
- **<td>** table data cell, one piece of data
- The number of cells in each **<tr>** determines the number of columns

Your First HTML! <img>

```
<!-- External Image -->
<img src="https://contents.bu.ac.th/contents/images/downloads/f4d1fd39-3034-4b71-b8e7-
a31b58ee5649.webp" alt="Profile" width="300" height="300">

<!-- Local Images -->
<img src="Images/BU_Logo.png" alt="Profile" width="300" height="300">
```

- **<img>** embeds an image, it has no closing tag
- **src** attribute the path to the image file
  External: a full URL from another server
  Local: a path inside your own project
- **alt** attribute a text shown if the image fails to load, and read aloud by screen readers
- **width** and **height** attribute: reserve space so the page does not jump while the image loads

HTML Tag Combination (1)

```html
<!-- Link inside a paragraph -->
<p>
    Bangkok University has operated since 1962. Visit the
    <a href="https://www.bu.ac.th/">official website</a>
    to learn more about our programs.
</p>

<!-- Links inside a list -->
<ul>
    <li><a href="https://www.bu.ac.th/">BU Website</a></li>
    <li><a href="https://www.w3schools.com/">W3Schools</a></li>
    <li><a href="https://html.spec.whatwg.org/">HTML Living Standard</a></li>
</ul>

<!-- Image inside a link, inside a paragraph -->
<p>
    <a href="https://www.bu.ac.th/">
        <img src="Images/BU_Logo.png" alt="Bangkok University homepage"
             width="150" height="150">
    </a>
</p>
```

HTML Tag Combination (2)

```html
<!-- Table with Images, Links, and Bold Text -->
<table>
    <tr>
        <th>Campus</th>
        <th>Preview</th>
        <th>Official Link</th>
    </tr>
    <tr>
        <td><b>Main Campus</b></td>
        <td>
            <img src="Images/Main_Campus.jpg" alt="BU Main Campus" width="100">
        </td>
        <td>
            <a href="https://www.bu.ac.th/en/campus" target="_blank">View Details</a>
        </td>
    </tr>
</table>
```

HTML Block Elements

- A block-level element always starts on a new line, and the browsers automatically add some space (a margin) before and after the element.
- A block-level element always takes up the full width available (stretches out to the left and right as far as it can).

```
<p>This is paragraph1</p>
<p>This is paragraph2</p>
<p>This is paragraph3</p>
<p>This is paragraph4</p>
<p>This is paragraph5</p>
```

Output

This is paragraph1

This is paragraph2

This is paragraph3

This is paragraph4

This is paragraph5

https://www.w3schools.com/html/html_blocks.asp

HTML Inline Elements

- An inline element does not start on a new line.
- An inline element only takes up as much width as necessary.

```
<a href="#">Anchor1</a>
<a href="#">Anchor2</a>
<a href="#">Anchor3</a>
<a href="#">Anchor4</a>
<a href="#">Anchor5</a>
```

Output



https://www.w3schools.com/html/html_blocks.asp

HTML Text Formatting

```
<!-- HTML text formatting tags -->
<p>This is normal text</p>
<p>This is <b>bold</b> text</p>
<p>This is <i>italic</i> text</p>
<p>This is <strong>strong</strong> text</p>
<p>This is <small>small</small> text</p>
<p>This is <sub>subscript</sub> text</p>
<p>This is <sup>superscript</sup> text</p>
<p>This is <ins>inserted</ins> text</p>
<p>This is <del>deleted</del> text</p>
<p>This is <mark>marked</mark> text</p>
```

Output

This is normal text

This is **bold** text

This is *italic* text

This is **strong** text

This is small text

This is subscript text

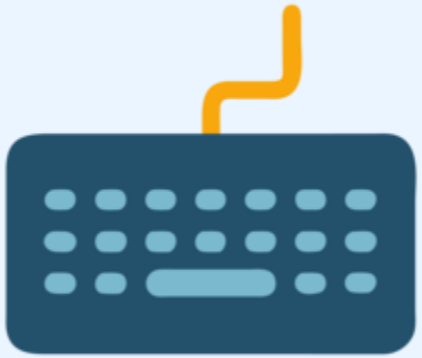
This is superscript text

This is inserted text

This is ~~deleted~~ text

This is marked text

https://www.w3schools.com/html/html_formatting.asp



HTML Form & Input Types

Check here: *https://www.w3schools.com/html/html_form_input_types.asp*

Form Element

```
<form target="_blank">
    <!-- Insert <input> elements here -->
    <button>Save</button>
</form>
```

Why we need <form> for input elements ?

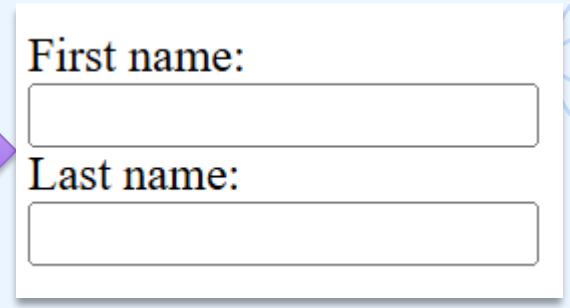
- **Data Packaging & Transmission:** The <form> tag acts as a container that bundles all child <input> data into a single payload, sending it to the backend via specified HTTP methods (GET or POST) to the destination endpoint (action).
- **Automatic Client-Side Validation:** Native browser checks (e.g., type="email", required, pattern) and constraint popups are **only** executed when submitting through an active <form>.
- **Keyboard Usability & Accessibility:** allowing users to submit data by pressing Enter inside any text input. Inside a <form>, any standard <button> automatically defaults to type="submit", instantly triggering submission without needing extra code.
- **Batch State Management:** Allows full form resets via <input type="reset"> and provides clean native integration for modern frontend APIs like FormData in JavaScript.

Label Tag on Input Element

Using the <label> tag to bind descriptive captions to specific <input> elements via the for attribute, improving accessibility for screen readers and increasing clickable hit areas for users.

```
<label for="fname">First name:</label><br>
<input type="text" id="fname" name="fname"><br>
<label for="lname">Last name:</label><br>
<input type="text" id="lname" name="lname"><br>
```

Output



The *for* attribute must insert the *id* of target element

https://www.w3schools.com/tags/tag_label.asp

Common Input Type (1)

Text & Data Entry

```
<!-- Text Input -->
<label>Username (text):</label><br>
<input type="text" name="username" placeholder="e.g. johndoe"><br><br>

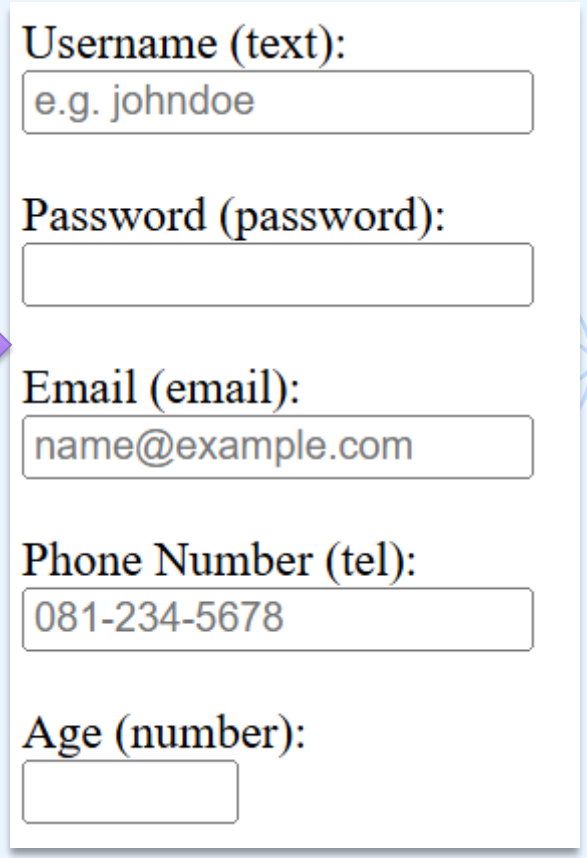
<!-- Password Input -->
<label>Password (password):</label><br>
<input type="password" name="password"><br><br>

<!-- Email Input -->
<label>Email (email):</label><br>
<input type="email" name="user_email" placeholder="name@example.com"><br><br>

<!-- Tel Input -->
<label>Phone Number (tel):</label><br>
<input type="tel" name="phone" placeholder="081-234-5678"><br><br>

<!-- Number Input -->
<label>Age (number):</label><br>
<input type="number" name="age" min="1" max="100"><br><br>
```

Output



Essential HTML Input Attributes

- **required:** Prevents form submission if the field is empty, triggering native browser error tooltips automatically.
- **placeholder:** Displays temporary hint text inside the field that disappears once the user begins typing.
- **value:** Sets the default data of the field or defines the exact payload value sent to the server for radios, checkboxes, and buttons.
- **disabled:** Disables user interaction completely, grays out the field, and excludes its data from being sent to the server.
- **readonly:** Prevents users from modifying the content while still allowing selection and submitting the data to the server.
- **min / max:** Defines the lowest and highest acceptable values for numeric (number) and calendar (date) inputs.
- **step:** Specifies the legal number intervals (e.g., step="0.01" or step="5") for numeric inputs.
- **checked:** Pre-selects a radio or checkbox input by default when the webpage loads.
- **maxlength:** Sets the maximum number of characters a user can enter into a text or password field.

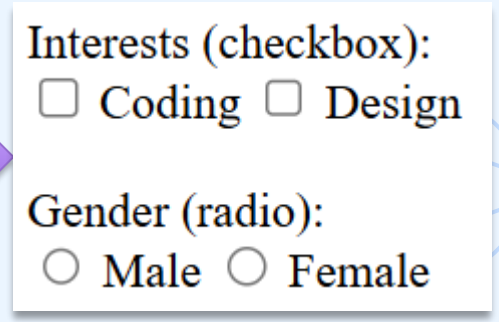
Common Input Type (2)

Choices & Selections

```
<!-- Checkboxes (Multiple Choice) -->
<label>Interests (checkbox):</label><br>
<input type="checkbox" name="interests" value="coding"> Coding
<input type="checkbox" name="interests" value="design"> Design<br><br>

<!-- Radio Buttons (Single Choice) -->
<label>Gender (radio):</label><br>
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female<br><br>
```

Output



Attributes

- **value (Actual Data Submitted):** Defines the specific data sent to the server when an option is selected; without it, the browser sends the generic default text "on" instead of meaningful data.
- **name for Radio Buttons (radio):** A shared name groups options together to enforce **single selection** (selecting one unchecks the others).
- **name for Checkboxes (checkbox):** A shared name groups related choices, submitting **all checked values** under that single category name.

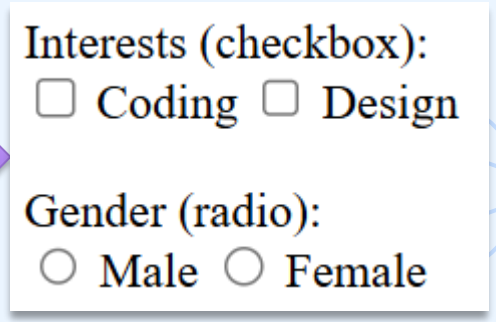
Common Input Type (2)

Choices & Selections

```
<!-- Checkboxes (Multiple Choice) -->
<label>Interests (checkbox):</label><br>
<input type="checkbox" name="interests" value="coding"> Coding
<input type="checkbox" name="interests" value="design"> Design<br><br>

<!-- Radio Buttons (Single Choice) -->
<label>Gender (radio):</label><br>
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female<br><br>
```

Output



Attributes

- **value (Actual Data Submitted):** Defines the specific data sent to the server when an option is selected; without it, the browser sends the generic default text "on" instead of meaningful data.
- **name for Radio Buttons (radio):** A shared name groups options together to enforce **single selection** (selecting one unchecks the others).
- **name for Checkboxes (checkbox):** A shared name groups related choices, submitting **all checked values** under that single category name.

Common Input Type (3)

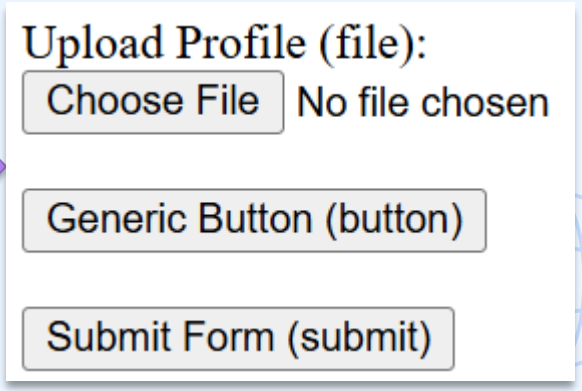
Actions & Files

```
<!-- File Input -->
<label>Upload Profile (file):</label><br>
<input type="file" name="avatar"><br><br>

<!-- Generic Button -->
<input type="button" value="Generic Button (button)"><br><br>

<!-- Submit Button -->
<input type="submit" value="Submit Form (submit)"><br><br>
```

Output



- **type="file":** Connects with backend storage and JavaScript to upload user avatars, parse data files, or preview images directly in the browser.
- **type="button":** Triggers interactive JavaScript functions to control dynamic UI elements, popups, and API requests without reloading the page.
- **type="submit":** Integrates with backend endpoints and asynchronous Fetch/AJAX requests to validate data and submit payloads seamlessly.

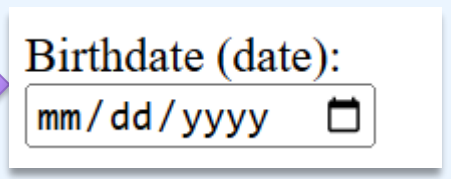
Common Input Type (4)

Date & Hidden Controls

```
<!-- Date Input -->
<label>Birthdate (date):</label><br>
<input type="date" name="birthdate"><br><br>

<!-- Hidden Input -->
<input type="hidden" name="csrf_token" value="abc123xyz">
```

Output

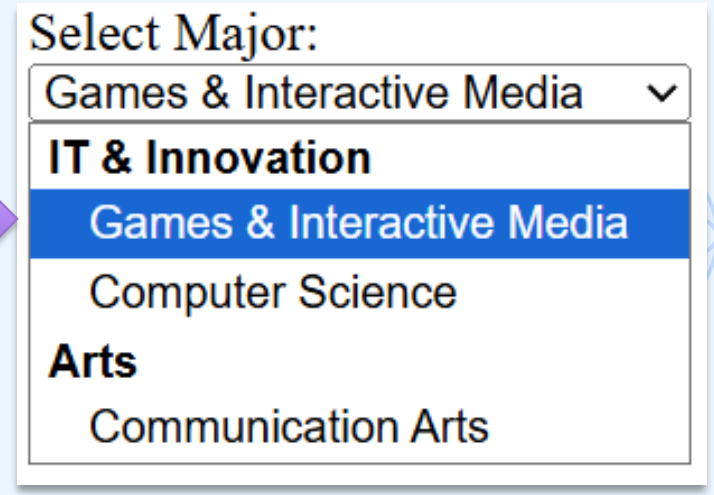


- **type="date" (Date Picker):** Provides a built-in calendar UI to capture standardized dates (YYYY-MM-DD), preventing input errors automatically.
- **type="hidden" (Hidden Data):** Stores background data (such as user IDs or security tokens) to send to the server without displaying it on the page.

Dropdown with Groups

```
<!-- Dropdown with Groups -->
<label for="major">Select Major:</label><br>
<select id="major" name="major">
  <optgroup label="IT & Innovation">
    <option value="games">Games & Interactive Media</option>
    <option value="cs">Computer Science</option>
  </optgroup>
  <optgroup label="Arts">
    <option value="comm_arts">Communication Arts</option>
  </optgroup>
</select><br><br>
```

Output

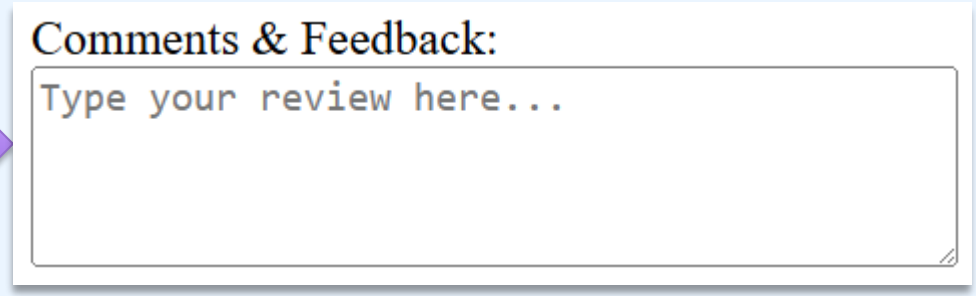


- **<optgroup> (Option Grouping):** Organizes related <option> items within a <select> dropdown into labeled, unselectable categories for clearer navigation in long lists.
- **<optgroup>** is completely ***optional***; simple dropdowns only require <select> and standalone <option> tags.

Multi-line Text Area

```
<label for="feedback">Comments & Feedback:</label><br>
<textarea id="feedback" name="feedback" rows="4" cols="40"
placeholder="Type your review here..."></textarea><br><br>
```

Output



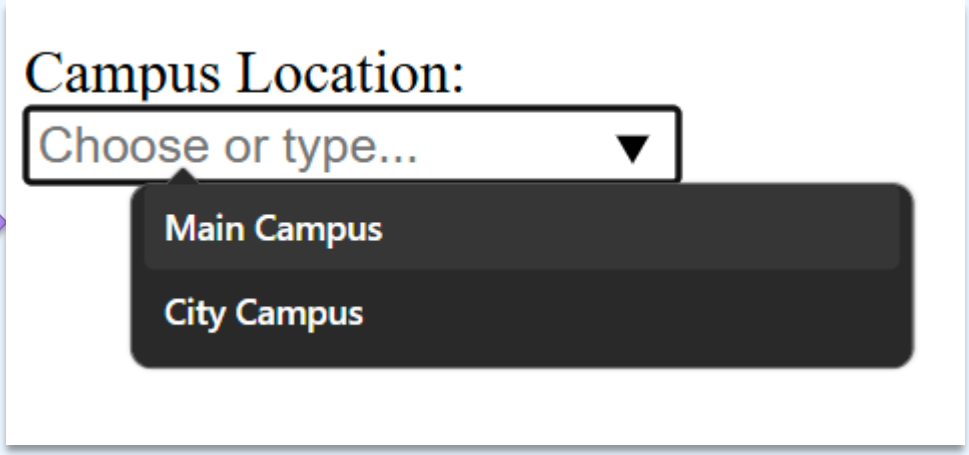
<textarea> essential attribute:

- **rows (Height):** Sets the visible number of text lines displayed vertically without needing to scroll.
- **cols (Width):** Sets the visible text width in average character spaces horizontally across the input area.

Autocomplete Datalist

```
<!-- Autocomplete Datalist -->
<label for="campus">Campus Location:</label><br>
<input list="campuses" id="campus" name="campus"
placeholder="Choose or type...">
<datalist id="campuses">
  <option value="Main Campus">
  <option value="City Campus">
</datalist><br><br>
```

Output

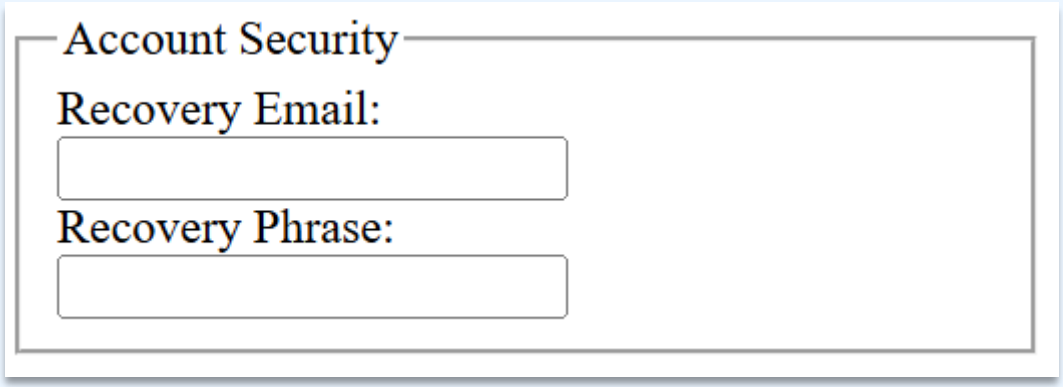


<datalist> (Autocomplete Suggestions): Provides a predefined list of recommended options for an <input> field, offering search-as-you-type suggestions while still allowing users to enter custom text.

Fieldset Grouping

```
<fieldset>
  <legend>Account Security</legend>
  <label>Recovery Email:</label><br>
  <input type="email" name="recovery_email"><br>
  <label>Recovery Phrase:</label><br>
  <input type="text" name="recovery_phrase">
</fieldset>
```

Output



<legend> (Caption/Title): Defines the caption or title for the <fieldset> border box, explaining what the grouped controls are for.

HTML INPUT TYPES

รวม Input Type ที่ใช้บ่อย พร้อมตัวอย่างการใช้งาน

ประเภท (Type)	</> โค้ดตัวอย่าง (Code Example)	ตัวอย่างการแสดงผล (Visual Output)	คำอธิบาย (Description)
Text ข้อความ	`<input type="text" />`	M-SoftTech	รับข้อมูลข้อความ 1 บรรทัด
Password รหัสผ่าน	`<input type="password" />`	•••••••••	รับรหัสผ่าน (ซ่อนตัวอักษร)
Number ตัวเลข	`<input type="number" />`	9	รับตัวเลขเท่านั้น
Checkbox ตัวเลือกหลายข้อ	`<input type="checkbox" />`	Option 1 Option 2	เลือกได้หลายตัวเลือก
Color เลือกสี	`<input type="color" />`		เลือกสี
Date เลือกวันที่	`<input type="date" />`	05 / 07 / 2026	เลือกวันที่
Time เลือกเวลา	`<input type="time" />`	10 : 30 AM	เลือกเวลา
Search ช่องค้นหา	`<input type="search" />`	Search Something	ช่องค้นหา
File อัปโหลดไฟล์	`<input type="file" />`	Choose File No file chosen	อัปโหลดไฟล์
Range ช่วงค่า (สไลเดอร์)	`<input type="range" />`		เลือกค่าจากช่วง (สไลเดอร์)
Radio ตัวเลือกเดียว	`<input type="radio" />`	Male Female	เลือกได้เพียง 1 ตัวเลือก
Submit ปุ่มส่งฟอร์ม	`<input type="submit" />`	Submit	ปุ่มส่งฟอร์ม
Week เลือกสัปดาห์และปี	`<input type="week" />`	Week, 05, 2026	เลือกสัปดาห์และปี

Tips & Best Practices

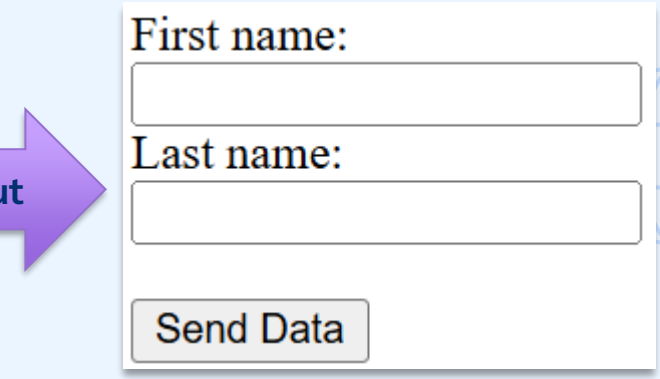
- ใช้ Input Type ให้เหมาะสม ช่วยให้ UX ดีขึ้น
- ตรวจสอบข้อมูลของผู้ใช้ ก่อนบันทึกเสมอ
- เพิ่มความสะดวก และความเร็วในการกรอกฟอร์ม
- ทำให้ฟอร์มตอบสนอง ใช้งานง่ายบนทุกอุปกรณ์
- HTML ง่ายแต่ทรงพลัง สร้างประสบการณ์ที่ดีได้!

Form Element Action Attribute

Using <form> tag collect user input by wrapping interactive elements (input tags) and submitting the captured data to a web server for processing.

```
<form action="" method="get">
    <label for="fname">First name:</label><br>
    <input type="text" id="fname" name="fname"><br>
    <label for="lname">Last name:</label><br>
    <input type="text" id="lname" name="lname"><br><br>
    <button>Send Data</button>
</form>
```

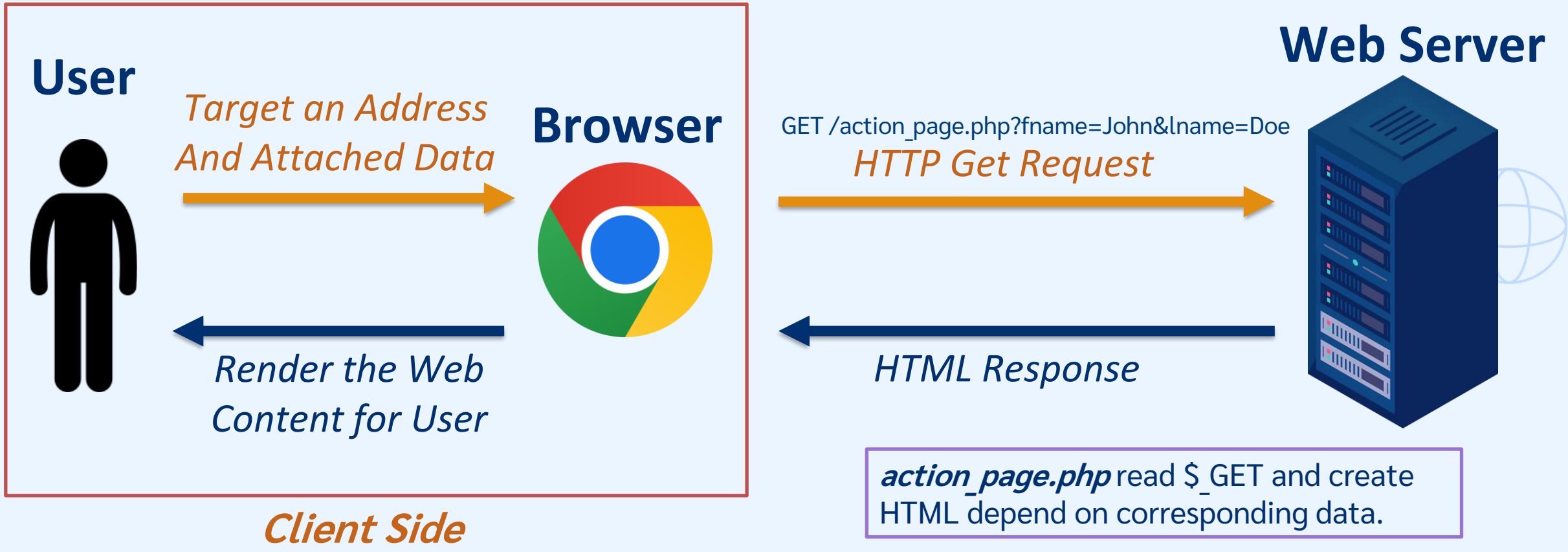
Output



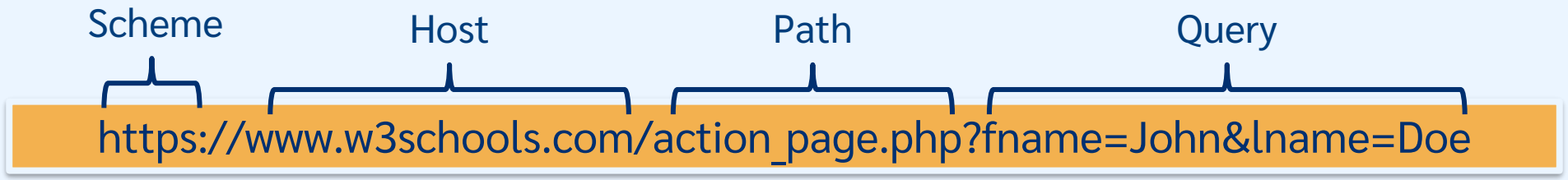
For <form> You can use Get and Post method to send a http request.

Try using GET method to send data (insert URL to *action* attribute) to: *https://www.w3schools.com/action_page.php*

What happens when we GET *action_page.php*?



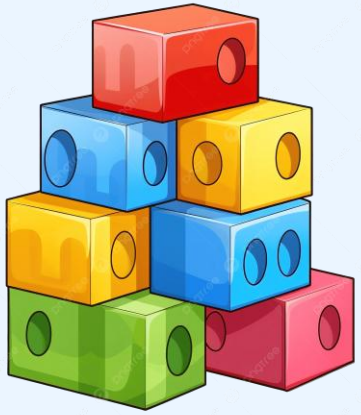
Result from GET *action_page.php*



Result URL

- **Scheme (https://):** Defines the protocol used for communication and encryption between the browser and the web server.
- **Host (www.w3schools.com):** The domain name identifying the specific web server that hosts the website.
- **Path (/action_page.php):** The exact location or file route on the server handling the incoming request.
- **Query (?fname=John&lname=Doe):** Form data appended as key-value pairs (name=value), separated by &, passed directly to the server via the GET method. The name of a query is <tag name="some-name"> that you define in HTML

If you want to see in-details data, try send to https://postman-echo.com/get



HTML Semantic Elements

Semantic Elements are HTML tags that clearly describe their meaning and the type of content they contain.

https://www.w3schools.com/html/html5_semantic_elements.asp

HTML5 Uses Division-Based Page Layouts

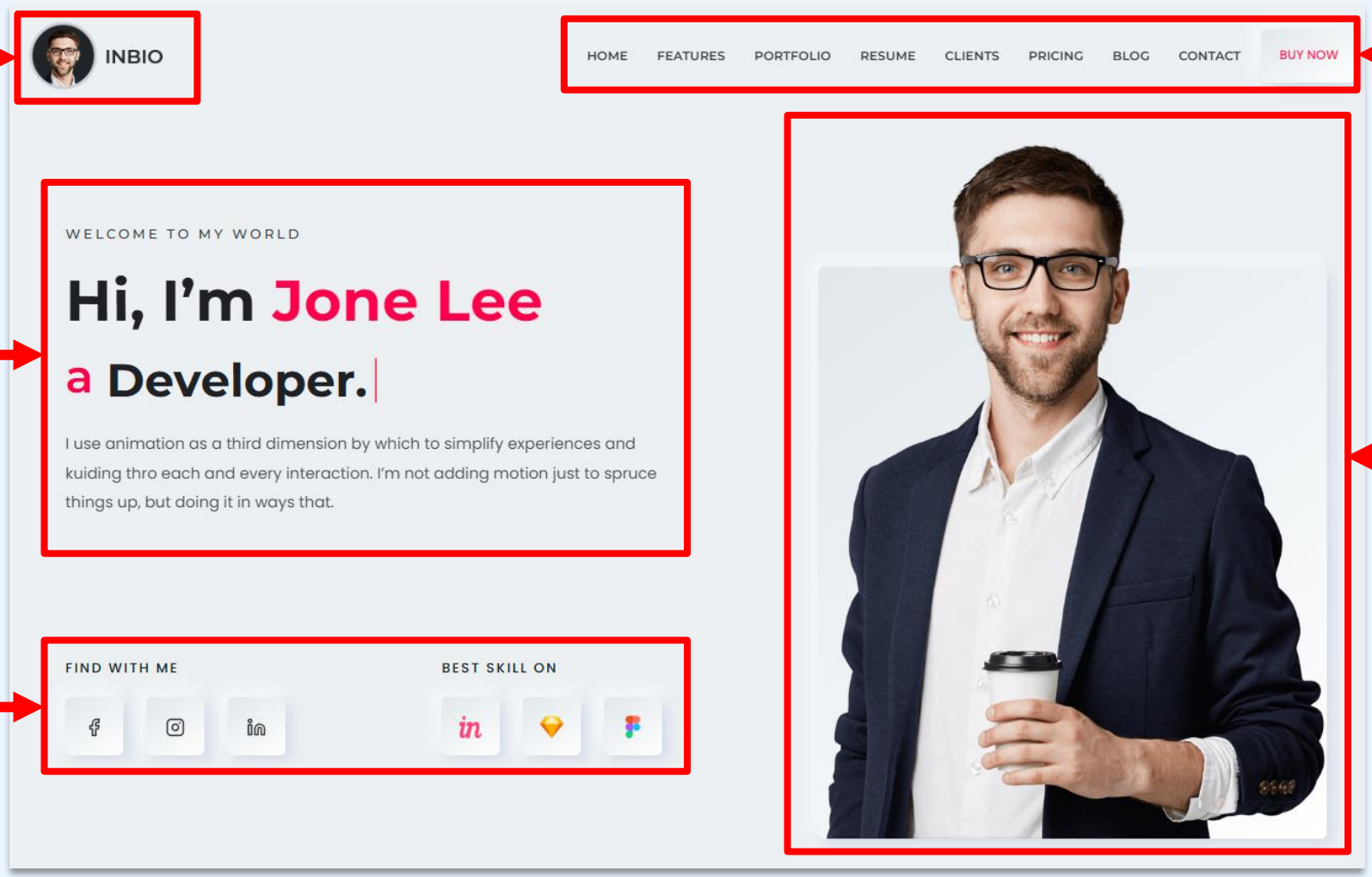
Brand/ Logo

Navigation bar & Action links

Headline & Bio intro

Hero image container

Social icons & Skill badges



https://rainbowit.net/html/inbio/index-white-version.html

Semantic Elements vs. Non-Semantic Elements

Semantic Element

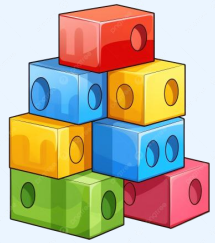
Tags that clearly describe their meaning and role in the document structure.

- **Example:** <header>, <nav>, <article>, <section>, <footer>, <main>, <aside>,
- **Enhanced Accessibility (a11y):** Browsers, search engines, and screen readers know exactly what type of content is inside.
- **Better SEO (Search Engine Optimization):** Helps web crawlers (like Google) index the most important content accurately.
- Self-documenting and easy for developer teams to read and maintain.

Non-Semantic Element

Generic containers used purely for layout and styling with no inherent meaning.

- **Examples:** ,
- Browsers treat them purely as generic boxes; engines cannot infer content purpose.
- No SEO value; search bots see generic structural containers.
- Often results in "div soup" (nested generic divs), making code harder to debug.
- Focus on styling purpose



Non-Semantic Element

```
<div class="product-grid">
    <div class="product-card">
        <h3>Ergonomic Mechanical Keyboard</h3>
        <div class="badge-discount">-20%</div>
        <p class="price">$89.99</p>
        <button type="button">Add to Cart</button>
    </div>
</div>
```

- **Core Purpose:** A neutral block-level container (divider) used strictly for layout, styling, and grouping elements without attaching semantic meaning.
- **Common Use Cases:** CSS Flexbox/Grid wrappers, styling container boundaries, visual badges, and layout columns.
- **Note:** By default, browsers always place a line break before and after the element.

Non-Semantic Element

```
<p>
    Special Offer: Get
    <span class="highlight-text">Free Shipping</span>
    on orders over
    <span class="price-tag">$50</span>!
</p>
<p>
    Status: <span class="badge-stock">In Stock</span> (Only 3 left)
</p>
```

- **Core Purpose:** An inline container used to target and style a specific portion of text or content without breaking onto a new line.
- **Common Use Cases:** Highlighting words, applying text colors, badge labels inside a sentence, or hooking small text fragments to JavaScript.
- **Key Rule:** Has no semantic meaning or line break behavior; it sits inline within parent block elements.

Semantic Element <section>

```
<section id="specs">
    <h2>Technical Specifications</h2>
    <p>Engineered for low latency, high durability, and seamless multi-device
switching.</p>
    <ul>
      <li><strong>Switch Type:</strong> Hot-swappable Mechanical</li>
      <li><strong>Connectivity:</strong> Wireless 2.4GHz, Bluetooth 5.3, USB-C</li>
      <li><strong>Battery Life:</strong> Up to 200 hours (RGB off)</li>
      <li><strong>Compatibility:</strong> Windows, macOS, Linux</li>
    </ul>
</section>
```

- **Core Purpose:** Groups standalone content that shares a single specific theme or topic within a document.
- **Common Use Cases:** Feature overviews, customer reviews, product specifications, contact areas, or FAQ sections on a landing page.
- **Key Rule:** A <section> should almost always contain its own heading (<h2>–<h6>) to clearly define the theme of that group.

Semantic Element <article>

```
<article draggable="true">
    <h2>Understanding Semantic HTML for Beginners</h2>
    <p><time datetime="2026-08-23">August 23, 2026</time> by <b>A. Arm</b></p>
    <p>
        Using semantic tags; makes your web pages cleaner, improves SEO, and
enhances accessibility for screen readers.
    </p>
    <a href="">Read full article...</a>
</article>
```

- **Common Use Cases:** Blog posts, news stories, forum posts, product cards, or user reviews.
- **Note:** The <article> element does not render as anything special in a browser. However, you can use CSS to style the element.

Semantic Element <nav>

```
<nav aria-label="Main Navigation">
    <ul>
      <li><a href="/">Home</a></li>
      <li><a href="products.html">Keyboards</a></li>
      <li><a href="deals.html">Special Offers</a></li>
      <li><a href="support.html">Contact Us</a></li>
    </ul>
</nav>
```

- **Core Purpose:** Declares a section containing major navigational links to help users and search engines navigate through the website or the current document.
- **Common Use Cases:** Primary website menus, breadcrumb trails, footer navigation bars, and table of contents.
- **Key Rule:** Reserved for **major navigation blocks only**; not all links on a page (like random links inside paragraphs) need to be inside a <nav>.

Semantic Element <header>

```
<header>
  <h1>TechGear Store</h1>
  <p>Your one-stop shop for mechanical keyboards and
accessories</p>
  <nav>
    <a href="products.html">Products</a> |
    <a href="deals.html">Deals</a> |
    <a href="support.html">Support</a>
  </nav>
</header>
```

- **Core Purpose:** Represents introductory content or a set of navigational aids for its parent container or the entire page.
- **Common Use Cases:** Website logos, branding, site-wide navigation bars, search inputs, or article titles with author metadata.
- **Key Rule:** You can have more than one <header> per page and one for the entire site (top of the page) and separate <header> tags inside <article> or <section> blocks.

Semantic Element <footer>

```
<footer>
    <p>&copy; <time datetime="2026">2026</time> TechGear Shop. All rights
reserved.</p>
    <p>
      Contact us: <a href="mailto:support@techgear.com">support@techgear.com</a>
    </p>
    <nav aria-label="Footer Legal Links">
        <a href="products.html">Products</a> |
        <a href="deals.html">Deals</a> |
        <a href="support.html">Support</a>
    </nav>
</footer>
```

- **Core Purpose:** Represents the concluding area for its nearest sectioning root or page, containing metadata, closing notes, or secondary navigational links.
- **Common Use Cases:** Copyright notices, terms of service, privacy policy links, sitemaps, author info, contact details, and social media icons.

Semantic Element <aside>

```
<aside style="font-size:larger; font-style:italic;
color:blue; float:right; width:120px;">
    <h3>Pro Tip</h3>
    <p>
      Lubing your mechanical switches significantly reduces
rattle and gives a smoother keystroke sound.
    </p>
</aside>
```

- **Core Purpose:** Holds secondary or supporting content that relates to the main topic but is separate enough that the page still makes full sense without it.
- **Common** Use Cases: Quick tips, callout boxes, "did you know" side notes, author bios, or related links.

Semantic Element <figure>

```
<figure>
    <img src="keyboard-switches.jpg" alt="Linear, tactile,
and clicky mechanical switches">
    <figcaption>Figure 1: Comparison of common mechanical
switch stem designs.</figcaption>
</figure>
```

- **Core Purpose:** Encapsulates self-contained media (like images, diagrams, charts, or code blocks) along with an explanatory caption.
- **Common Use Cases:** Product photos, technical diagrams, flowcharts, infographics, and code examples.
- **Key Rule:** Pair with <figcaption> (as the first or last element inside <figure>) to provide an accessible, semantic label for the media.



Browser Developer Tools (F12)

A suite of built-in web authoring and debugging tools integrated directly into every modern browser (opened via the F12 shortcut or Right-Click > Inspect).

- **Elements Tab (DOM & Styles):** Inspect and live-edit HTML tags, attributes, and CSS rules in real time to preview design changes instantly.
- **Console Tab (Debugging):** View JavaScript runtime errors, warnings, and log statements (console.log), and execute quick JavaScript commands directly in the browser.
- **Network Tab (Traffic & Forms):** Monitor outgoing HTTP requests (like GET and POST form submissions), view payload parameters, and inspect server responses and status codes.

Check This Week Assignment on

MS Team

Assignment Example

Natdanai Computer | Personal Portfolio

นักศึกษาวิทยาการคอมพิวเตอร์ ผู้สนใจในการพัฒนา **Frontend Web Development** และ **Software Engineering**

- ประวัติส่วนตัว
- ประวัติการศึกษา
- ทักษะและความสนใจ
- แบบฟอร์มติดต่อ

ประวัติส่วนตัว (About Me)



สวัสดีครับ! ผมชื่อ **นายณัฐดนัย คอมพิวเตอร์** ปัจจุบันกำลังศึกษาอยู่ชั้นปีที่ 3 สาขาวิชาวิทยาการคอมพิวเตอร์ คณะเทคโนโลยีสารสนเทศและนวัตกรรม มหาวิทยาลัยกรุงเทพ

ผมมีความหลงใหลในการสร้างสรรค์เว็บแอปพลิเคชันที่ใช้งานง่าย และกำลังมุ่งมั่นเรียนรู้ในรายวิชา CS319 Internet Programming I เพื่อพัฒนาทักษะสู่การเป็น *Full-Stack Developer* มืออาชีพในอนาคตครับ

ประวัติการศึกษาและผลงาน (Education & Experience)

ปีการศึกษา	ระดับการศึกษา / สถาบัน	สาขาวิชา / เกรดเฉลี่ย
2024 - ปัจจุบัน	ปริญญาตรี - มหาวิทยาลัยกรุงเทพ	วิทยาการคอมพิวเตอร์ (CS)
2021 - 2023	มัธยมศึกษาตอนปลาย - โรงเรียนวิทยาศาสตร์	แผนการเรียน วิทย์-คณิต (GPA: 3.85)

ทักษะและความสนใจ (Skills & Interests)

ทักษะทางเทคนิค (Technical Skills)

1. **HTML5 & CSS3:** การจัดโครงสร้าง Semantic Web และการจัดวาง Layout
2. **Programming Languages:** C++, Java, และ JavaScript (ES6+)

ทักษะและความสนใจ (Skills & Interests)

ทักษะทางเทคนิค (Technical Skills)

1. **HTML5 & CSS3:** การจัดโครงสร้าง Semantic Web และการจัดวาง Layout
2. **Programming Languages:** C++, Java, และ JavaScript (ES6+)

3. **Developer Tools:** VS Code, Git, GitHub, และ Chrome DevTools

งานอดิเรกและความสนใจ (Hobbies & Interests)

- ติดตามข่าวสารเทคโนโลยีเว็บ และ AI Tools สมัยใหม่
- ฝึกแก้ปัญหาโจทย์อัลกอริทึมบน LeetCode
- เล่นดนตรี และฟังเพลงเพื่อผ่อนคลาย

ติดต่อฉัน (Contact Me)

หากอาจารย์หรือผู้ที่สนใจต้องการสอบถามข้อมูลเพิ่มเติม สามารถส่งข้อความถึงผมได้ผ่านแบบฟอร์มนี้ครับ:

แบบฟอร์มส่งข้อความติดต่อ

ชื่อ-นามสกุล ผู้ติดต่อ:
กรุณากรอกชื่อของคุณ

อีเมลตอบกลับ:
example@bu.ac.th

หัวข้อการติดต่อ:
สอบถามข้อมูลทั่วไป

ข้อความเพิ่มเติม:
พิมพ์ข้อความของคุณที่นี่. . .

ส่งข้อความ (Submit) ล้างข้อมูล (Reset)

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Thank you!